

**B. Sc. (Hons.) Semester - V Examination, 2018-19****PHYSICS****Paper : BPT-501  
(Mathematical Physics)****Time : Three Hours****[ Full Marks : 70****[Write your Roll No. at the top immediately on the receipt of this question paper.]****Note :** Attempt all questions. Marks are indicated in the right margin.**1. Answer any six of the following :****6 × 3 = 18**

- (a) What are Hermitian matrices? Explain using example.
- (b) Show that Kronecker  $\delta$  is a mixed tensor.
- (c) Prove that outer product of two vectors is a tensor.
- (d) Find the condition for which a second order linear differential equation reduces to a first order equation.
- (e) Find the values of first four members of  $L_n(x)$ .
- (f) What is the need of scale factor for transformation from Cartesian to polar coordinates?
- (g) Resolve Cartesian unit vectors into their circular cylindrical components and hence obtain expression for  $\vec{r}$ .

**P. T. O.**

2. (a) Obtain an expression for  $\nabla^2\psi$  in spherical polar coordinates. **5**
- (b) Write the characteristic equation of a matrix and there by show that  $A^3 = 3A^2 - A + 3$  for

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

**8****OR**

- (a) Show that for a central force  $f(r)$ ;

$$\vec{\nabla}^2 f(r) = \frac{2}{r} f'(r) + \frac{\partial^2 f(r)}{\partial r^2}$$

**5**

(b) For the matrix  $A = \begin{pmatrix} 3 & 1 & -1 \\ -2 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix}$  find the eigen values and hence diagonalize the matrix. **8**

3. (a) Prove that  $nP_n(x) = xP'_n(x) - P'_{n-1}(x)$ . **5**

(b) Find the eigen values and eigen vectors of

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

**8**

( 2 )

OR

- (a) Show that  $\exp[2xt - t^2]$  is the generating function of Hermite's polynomial  $H_n(x)$ . **5**
- (b) Find the Fourier series for the function  $f(x) = x^2$ ;  $-\pi < x < \pi$  and hence show that  $\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}$ . **8**
4. (a) Prove that  $\int_0^1 x J_n(\alpha x) \cdot J_n(\beta x) dx = 0$  for  $\alpha \neq \beta$ , where  $\alpha$  and  $\beta$  are two roots of  $J_n(x) = 0$ . **5**
- (b) For a periodic function defined as

$$f(x) = \begin{cases} -1; & -\pi < x < 0 \\ 1; & 0 < x < \pi \end{cases} \quad \text{find the Fourier series.}$$

**8**

5. (a) If for a linear oscillator,  $y = \sin x$  is one solution, find its second solution. Also, show that the two solutions are linearly independent. **5**
- (b) Find the Fourier series

$$f(x) = \begin{cases} \pi x, & 0 \leq x \leq 1 \\ \pi(2 - x) & 1 \leq x \leq 2 \end{cases}$$

**8**

— o —

---

#### External Portals & Academic Resources:

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app>